Frequency Based Ranging (FBR) for Bluetooth Channel Sounding on a Single 2 MHz Channel

Santosh Nagaraj
San Diego State University
San Diego, California, USA
snagaraj@sdsu.edu

Abstract

In this paper, a novel Frequency Based Ranging (FBR) technique to locate Bluetooth devices is shown and studied. It achieves the accuracy of Phase Based Ranging (PBR) methods with a much smaller bandwidth and in less time. FBR will allow decimeter level ranging accuracy from packet exchange on a single 2 MHz Bluetooth channel. Like PBR, FBR is also robust to multipath effects. FBR is based on the basic principle of Frequency Modulated Continuous Wave (FMCW) radars. However, classical linear chirp FMCW is not suitable for ranging Bluetooth devices for a variety of reasons that also include waveform incompatibility. In this paper, a modified form of the FMCW radar is shown to be compatible with Bluetooth. A simple way to generate the waveform within an unmodified Bluetooth packet structure is also shown. Signal processing steps to estimate the target range are compatible with widely used Bluetooth receiver technologies. Simulations over indoor channels with exponential power-delay profiles confirm the ranging accuracy of the FBR method. When this ranging method is combined with Bluetooth direction finding, accurate indoor localization will be possible from a single packet exchange.

CCS Concepts

• **Hardware → Digital signal processing**.

Keywords

Internet of Things, Bluetooth Channel Sounding, Frequency Modulated Continuous Wave (FMCW).

ACM Reference Format:

Santosh Nagaraj. 2025. Frequency Based Ranging (FBR) for Bluetooth Channel Sounding on a Single 2 MHz Channel. In *The 15th International Conference on the Internet of Things (IOT 2025), November 18–21, 2025, Vienna, Austria.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3770501.3770519

1 Introduction

Bluetooth Low Energy (BLE) [22] is a widely used communication protocol for the transmission of relatively small amounts of data over short distances in an efficient manner. The standard is also feasible for implementation in batteryless tags owing to its simplicity [13, 16]. While Bluetooth was initially used only for communication, it is currently being used for localization as well. Its low hardware and operational cost has made BLE the communication and localization technology of choice for devices connected to the Internet of Things (IoT) [4, 17, 18]. In Bluetooth 5.1 Channel Sounding [22], an *Initiator* transmits the RF signal that is received, processed and retransmitted by the tag *Reflector* to facilitate ranging. Another new feature of the Bluetooth 5.1 standard is direction-finding, wherein, multiple antennas on either the transmitting or the receiving BLE device can be leveraged for Angle of Departure (AoD) or Angle of Arrival (AoD) estimation. When combined with ranging, the location of the device can be identified [3]. Localization can be leveraged in practical applications such as inventory monitoring, target tracking, etc. Bluetooth 5.1 introduced the Constant Tone Extension (CTE) feature specifically for the purpose of channel sounding [23].

In the context of BLE channel sounding, three methods have been extensively studied in the literature: (i) Received Signal Strength Indicator (RSSI) values can be used to approximate the distance of a transmitter from the receiver [12, 20]. However, the accuracy of RSSI based sounding is low in indoor environments owing to fading. Researchers have improved the accuracy by incorporating fingerprinting [7, 8], wherein the RSSI values at different locations are measured and stored for comparison with the RSSI value in a test position. (ii) Phase Based Ranging (PBR) [2, 25] is a BLE recommended method specifically for ranging compatible backscatter tags. In this method, the difference between the carrier phase transmitted by the initiator and that of the backscattered signal is measured to estimate the two-way distance. The ambiguity introduced by the wrapping of the phase angle can be mitigated by taking the measurements at a multiple carrier frequencies. Some authors suggest measurements across the entire 80 MHz Bluetooth spectrum. (iii) A more recent method is the Round Trip Time (RTT) [10, 11] between the transmission and reception of an encrypted and scrambled BLE packet. This method is not very accurate but provides a secure method for bounding the range. In general, all ranging methods are sensitive to inaccuracies and instabilities in the internal carriers or clocks of the devices involved and to multipath effects.

One ranging method that has hitherto not been used in Bluetooth channel sounding is based on Frequency Modulated Continuous Wave (FMCW) [6, 21, 24] radars. With FMCW radars, propagation delays are converted to beat frequencies that can be detected using a Fast Fourier Transform (FFT) or other techniques. As is also the case with PBR, flat-fading multipath does not directly impact measurements since propagation delays are independent of the signal amplitude. However, the Signal to Noise Ratio (SNR) of the range estimate depends on the received signal amplitude. This suggests excellent ranging ability over indoor wireless channels

IOT 2025, Vienna, Austria

ACM ISBN 979-8-4007-1595-2/25/11
https://doi.org/10.1145/3770501.3770519

which are known to exhibit flat fading over the narrow 2 MHz bandwidth of a single BLE channel.

In this paper, a Frequency Based Ranging (FBR) method for ranging is proposed and studied for Bluetooth devices. *It is shown that centimeter level accuracy can be obtained with just 2 MHz of bandwidth (a single Bluetooth channel) in a matter of a few milliseconds.* The proposed FBR technique is based on the FMCW radar method and is adapted for implementation in unmodified Bluetooth hardware. It is shown that the technique can be implemented in a way that all transmitted signals are compatible with the Bluetooth standard. Beat frequencies are calculated from the Inphase and Quadrature (I/Q) samples obtained in commonly used BLE demodulation techniques such as the frequency discriminator or quadri-correlator. This paper studies BLE-compatible transmit-signal generation and receive-signal processing steps to estimate the propagation delay. When coupled with AoA or AoD direction-finding, FBR allows tag localization from a single Bluetooth 5.1 packet.

The remainder of this paper is organized as follows. In Section 2, literature pertaining to the FMCW ranging method is discussed. Mathematical principles of FMCW based ranging is also summarized. In Section 3, the proposed Bluetooth device ranging technique is described. In Section 4, simulation results are shown to demonstrate ranging performance and accuracy. Lastly, Section 5 concludes the paper with observations and notes on future work.

2 Background & Related Works

Measuring propagation delays directly is impractical in many applications since very wide bandwidths are required. FMCW is a ranging method wherein propagation delays are converted to frequency differences [21, 24]. An FMCW radar transmits a continuous wave with a frequency is that varying with time. In other words, the radar transmits a frequency modulated waveform. Propagation delays result in instantaneous frequency differences. In general, it is much easier to measure frequency differences than to directly measure delays. In some ways, FMCW is similar to PBR, which is based on the fact that measuring phase differences is easier than measuring time delays. Some commonly used modulating waveforms for FMCW are linear, triangular and sinusoidal.

In Linear FMCW (L-FMCW), the modulating waveform is a periodic sawtooth waveform that sweeps a bandwidth B over a time T. Propagation delay of t_d seconds appear as frequency shifts of Bt_d/T, which is generally measured by an FFT (called the Range-FFT in literature). A second FFT, called the Doppler-FFT, processes the signal across multiple periods of the sawtooth waveform to track motion. For example, L-FMCW radars with 5 GHz chirp bandwidth are widely used in automotive systems such as Adaptive Cruise Control owing to their ability to resolve multiple closely-spaced delays (or targets). When confined to the bandwidth B, the best range accuracy possible with L-FMCW is $c/2B$ meters, where c is the speed of light. This is due to the FFT bin width limitation for a given B and T. With a BLE channel bandwidth of 2 MHz, the device can only be located within a 75 meter accuracy. This method is therefore not suitable for Bluetooth channel sounding.

An FMCW variant that has seen only limited applications so far is Sinusoidal-FMCW (S-FMCW) [19, 26]. In this method, the modulating waveform for FMCW is a sinusoid with no natural coherent processing intervals such as the sawtooth end-points in L-FMCW. The continuous variation of the instantaneous frequency allows for longer processing durations and consequently results in improved ranging accuracy even with narrow bandwidths. S-FMCW has been used in optical wavelength ranging applications [26] and to measure small movements of the target [19]. However, the ability of the L-FMCW radar to resolve multipath delays is lost. Resolving multiple delays is necessary for imaging and other tasks involving multiple targets. However, it is not necessary for ranging a single target. We believe that S-FMCW is ideal for Bluetooth Channel Sounding since devices are in connection mode and there is only one distance to range.

2.1 Sinusoidal FMCW Ranging Principle

An S-FMCW radar transmits a sinusoidally modulated FM signal $g(t)$ defined by the complex envelope:

$$\begin{aligned} g(t) &= e^{j\phi(t)}, \text{ where,} \\ \phi(t) &= \omega_c t + 2\pi A \int_0^t \sin(\omega_m \tau)\, d\tau. \end{aligned} \tag{1}$$

Here, $\phi(t)$ is the phase function of $g(t)$; ω_c is the carrier frequency; $\omega_m = 2\pi f_m$ and A are the frequency and amplitude of frequency modulation, respectively. Bandwidth of $g(t)$ can be calculated from Carson's Rule [9, 15] as follows:

$$B_{s-fmcw} = 2(A + f_m). \tag{2}$$

Suppose that $g(t)$ is reflected and received with a round-trip propagation delay t_d as follows:

$$\begin{aligned} x(t) &= h\, e^{j\phi'(t)}, \text{ where,} \\ \phi'(t) &= \omega_c(t - t_d) + 2\pi A \int_0^{t-t_d} \sin(\omega_m \tau)\, d\tau, \end{aligned} \tag{3}$$

and h is the flat-fading channel gain. The radar correlates the transmitted and received signals to get:

$$\begin{aligned} r(t) &= g(t) \times x^*(t) \\ &= h\, e^{j\theta(t)}, \text{ where,} \\ \theta(t) &= \omega_c t_d + 2\pi A \int_{t-t_d}^{t} \sin(\omega_m \tau)\, d\tau. \end{aligned} \tag{4}$$

The instantaneous frequency of the correlator output is:

$$\begin{aligned} y(t) &= \frac{1}{2\pi} \times \frac{d}{dt}\theta(t) \\ &= A\sin(\omega_m t) - A\sin(\omega_m t - \omega_m t_d) \\ &= 2A \sin\left(\frac{\omega_m t_d}{2}\right) \cos\left(\omega_m t - \frac{\omega_m t_d}{2}\right) \\ &\approx A\omega_m t_d \cos\left(\omega_m t - \frac{\omega_m t_d}{2}\right). \end{aligned} \tag{5}$$

The last equation is because f_m will be chosen in the sub-MHz range while t_d will be under a few tens of nanoseconds. Equation (5) shows that the beat frequency $A\omega_m t_d$ is proportional to the round-trip time-of-flight between the transmitter and the receiver and, thus, enables ranging. The beat frequency can be measured in a few different ways. In [26], a solution based on measuring the intensity (amplitude) of the correlator output has been studied. In [19], a method based on Fourier Series decomposition of the output

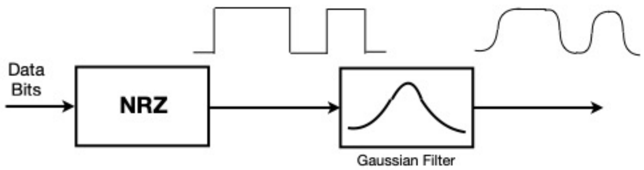

Figure 1: Block diagram of GFSK baseband. The output feeds an FSK modulator.

signal has been shown. Alternatively, direct frequency demodulation of the correlator output $r(t)$ can yield the signal $y(t)$. This FM demodulation approach is most compatible with Bluetooth radios since frequency discrimination is an integral aspect of BLE signal demodulation [14].

2.2 Bluetooth Low Energy

Bluetooth operates in the unlicensed spectrum between 2.400–2.483 GHz. Bluetooth spectrum is divided into 40 channels of 2 MHz width each. Three of these channels are reserved for connectionless advertisements while the rest are used for data connections. In connection-oriented modes, Bluetooth devices use frequency hopping to avoid deep fades and improve reliability. In its basic Low Energy mode, information is packetized and transmitted at the rate of 1 Mbps. Data may or may not be encoded with a convolutional code depending on the packet type. Packet durations range from a few hundred microseconds to over a millisecond.

Figure 1 shows the block diagram of a Bluetooth transmitter implementing Gaussian-filtered Frequency Shift Keying (GFSK) modulation. Data bits are first represented by a bipolar Non Return to Zero (NRZ) line code $d_i \in \{-1, +1\}$. The resulting baseband signal is next filtered by a Gaussian impulse response with time-bandwidth product $BT_b = 0.5$, where $T_b = 1\ \mu$s is the bit duration. The Gaussian filter serves to smoothen the bit transitions and limits the baseband bandwidth to 500 kHz. The filtered baseband signal is lastly frequency modulated with a nominal frequency deviation $\Delta f = 250$ kHz/V.

3 Bluetooth Frequency Based Ranging (FBR)

It is possible to approximate the S-FMCW waveform of (1) within the limitations of the Bluetooth standard. It is a matter of identifying a periodic sequence of data bit 1s and 0s that results in a nearly-sinusoidal baseband waveform. We call this sequence the Alternating Tone Sequence (ATS). When the ATS is FSK modulated in the final step of Bluetooth transmission, the resulting signal will resemble an S-FMCW waveform. The ATS can be included in the data segment of the packet itself, or as an extension analogous to the CTE of Bluetooth 5.1 standard. We consider two candidate sequences for this: ATS 1: sequence of alternating 1s and 0s, i.e., 10101010 . . ., which has the fundamental frequency $f_m = 500$ kHz and ATS 2: sequence of two 1s and two 0s, i.e., 110011001100 . . ., which has the fundamental frequency $f_m = 250$ kHz. Figure 2 shows the NRZ waveform and the corresponding Gaussian filtered waveform for ATS 1. Table 1 shows the Fourier Series coefficients, c_k of the Gaussian filtered ATS 1 and 2.

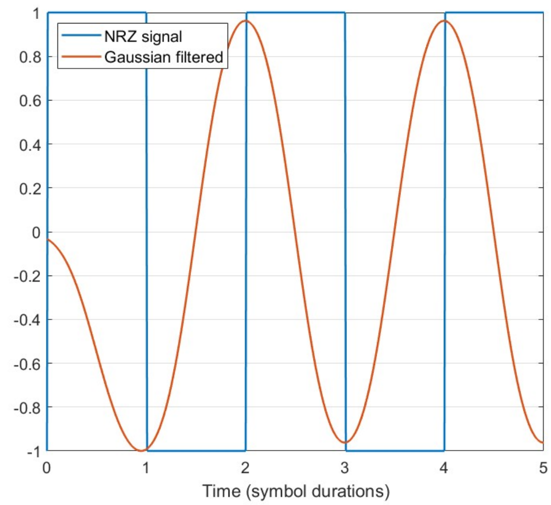

Figure 2: NRZ waveform and Gaussian-filtered waveform for ATS sequence 1.

·	c_1	c_2	c_3	c_4
$f_m = 500$ kHz	-6 dB	-20 dB	-39 dB	-60 dB
$f_m = 250$ kHz	-2 dB	-10 dB	-19 dB	-34 dB

Table 1: Fourier Series coefficients for ATS sequences 1 & 2

From Table 1, it appears that ATS 1 is a better sinusoid as Fourier Series coefficient c_1 is at least 14 dB above the others. For ATS 2, c_1 is only 8 dB above the next one. However, relative values of the coefficients alone do not convey the full picture. Coefficient c_1 of ATS 1 is 4 dB below the corresponding coefficient of ATS 2. This can be attributed to the Gaussian filter, which has a 500 kHz bandwidth. ATS 2 has its fundamental frequency at 250 kHz which is not attenuated as much as the fundamental frequency of ATS 1, which is at 500 kHz.

A second aspect of interest is the spectrum of the resulting GFSK waveform. Figure 3 shows the spectrum of GFSK modulated ATS 1 at the carrier frequency $f_c = 2.45$ GHz. It can be seen that the RF spectrum is only 1 MHz wide while the Bluetooth standard allows up to 2 MHz even for the LE mode. Outside the 1 MHz band, all spectral lines are at least 30 dB below the peak. The narrow bandwidth compared to a typical GFSK signal can be attributed to the periodicity of the ATS 1 sequence. The bandwidth is even lower for the ATS 2 sequence. This suggests that the frequency deviation Δf can be increased to 500 kHz without increasing the bandwidth beyond 2 MHz. This is possible with existing Bluetooth hardware too. In fact, the LE-2M mode introduced by the Bluetooth 5.1 standard uses $\Delta f = 500$ kHz.

3.1 Channel Sounding

In Bluetooth Channel Sounding, two devices— an Initiator and a Reflector— first establish a connection. This will allow the devices

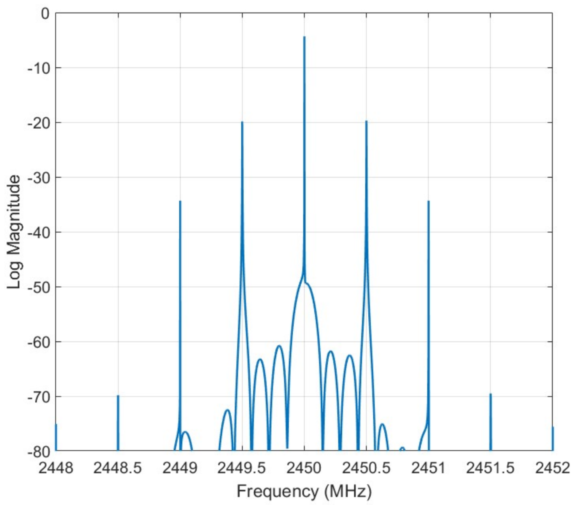

Figure 3: RF spectrum for GFSK modulated ATS 1 at $f_c = 2450$ MHz. The 30 dB attenuation bandwidth is only 1 MHz.

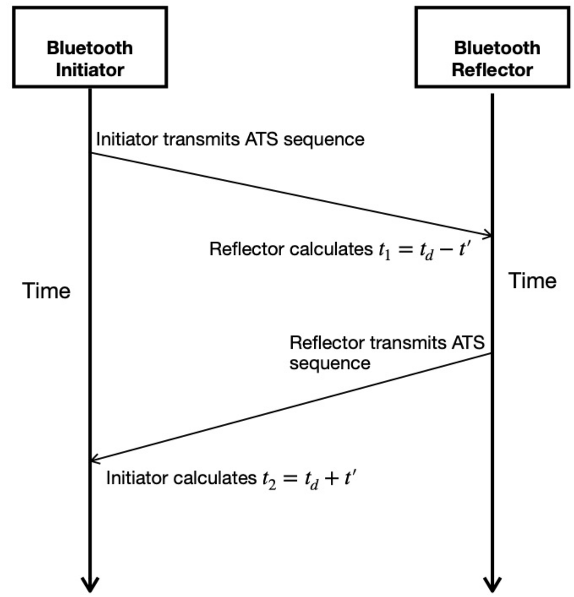

Figure 4: FBR involves one transmission from Initiator (Bluetooth master) to Reflector (Bluetooth client) and one from Reflector to Initiator.

to perform coarse carrier frequency and symbol timing corrections. Analogous to existing PBR techniques in the literature, FBR will also involve two transmission phases— one from Initiator to Reflector and one from Reflector to Initiator. The Initiator sends a connection request to the Reflector, thus becoming the master unit of the connection. The Reflector assumes the role of a Bluetooth client in the data connection. The Bluetooth standard requires transmission from master to client be immediately followed by a transmission from client to master. The FBR channel sounding algorithm is explained next. Signal processing steps differ from those of the S-FMCW radar in the Section 2.1 due to the transmit and receive operations being performed in different devices with the possibility of frequency and timing offsets between them.

Phase 1: Initiator to Reflector. The Initiator transmits a data packet that includes an ATS component $g(t)$, of a certain duration T, in a predetermined part of the packet. The Reflector receives the packet and extracts the ATS component in I/Q format from the packet. This received ATS component can still be represented as $x(t) = hg(t-t_d)$ as in (3). At the same time, the receiver constructs a copy, $\tilde{g}(t)$, of the S-FMCW signal as follows:

$$\begin{aligned}\tilde{g}(t) &= e^{j\tilde{\phi}(t)}, \text{ where,}\\ \tilde{\phi}(t) &= (\omega_c - \omega')(t - t') + 2\pi A \int_0^{t-t'} \sin(\omega_m \tau)\, d\tau, \end{aligned} \tag{6}$$

where, ω' and t' represent the carrier frequency mismatch and symbol timing mismatch, respectively. So, the Reflector's frequency and time are assumed to be $\omega_c - \omega'$ and $t - t'$, respectively. We assume that t' and ω' do not change over the duration of the channel sounding process. The receiver correlates the received signal $x(t)$ with the prototype $\tilde{g}(t)$ to get $r(t)$ as in Section 2.1:

$$\begin{aligned} r(t) &= \tilde{g}(t) \times x^*(t)\\ &= h\, e^{j\theta(t)}, \text{ where,}\\ \theta(t) &= \theta_0 - \omega' t + 2\pi A \int_{t-t_d}^{t-t'} \sin(\omega_m \tau)\, d\tau. \end{aligned} \tag{7}$$

In the above equation, θ_0 is a constant. Instantaneous frequency of the correlator output is:

$$\begin{aligned} y_1(t) &= \frac{1}{2\pi} \times \frac{d}{dt}\theta(t)\\ &= -\frac{\omega'}{2\pi} + A\sin(\omega_m(t-t')) - A\sin(\omega_m(t-t_d))\\ &= -\frac{\omega'}{2\pi} + 2A\sin\left(\frac{\omega_m(t_d-t')}{2}\right)\cos\left(\omega_m t - \frac{\omega_m(t_d+t')}{2}\right)\\ &\approx -\frac{\omega'}{2\pi} + A\omega_m(t_d-t')\cos\left(\omega_m t - \frac{\omega_m t_d + \omega_m t'}{2}\right). \end{aligned} \tag{8}$$

The approximation in the last equation still holds as both t_d and t' can be expected to be in the nanosecond range while f_m is 500 kHz at most. The Reflector will then correlate $y_1(t)$ with $\cos(\omega_m(t-t'))$ and $\sin(\omega_m(t-t'))$ over the observation duration T to get $t_{1,c}$ and $t_{1,s}$ as follows:

$$\begin{aligned} t_{1,c} &= \frac{2}{T}\int_{t'}^{T+t'} y_1(t)\ \cos(\omega_m(t-t'))\, dt\\ t_{1,s} &= \frac{2}{T}\int_{t'}^{T+t'} y_1(t)\ \sin(\omega_m(t-t'))\, dt. \end{aligned} \tag{9}$$

It can be seen from (8) that $t_{1,c}$ and $t_{1,s}$ will allow the estimation of $t_1 = t_d - t'$.

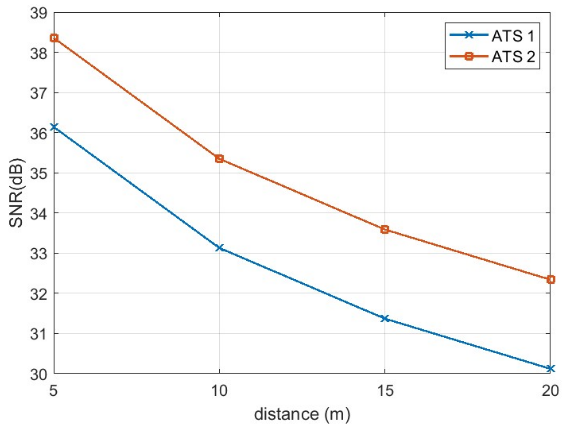

Figure 5: SNR of the range estimate at different distances.

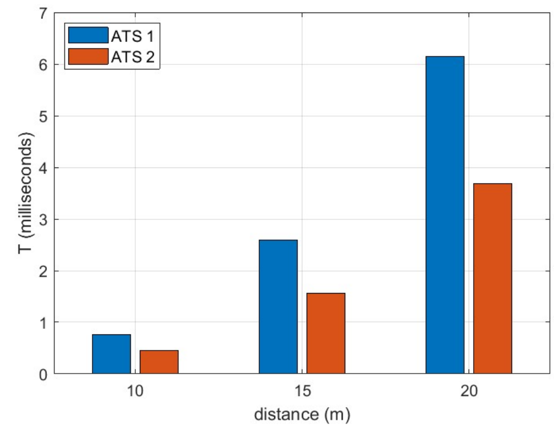

Figure 6: ATS duration necessary for 10 cm accuracy at different distances.

Phase 2: Reflector to Initiator. The Reflector will next transmit a data packet to the Initiator that will include the ATS sequence while the Initiator processes the received signal in exactly the same way as before. The only difference is that the relative frequency and timing offsets are now reversed, i.e., $-\omega'$ and $-t'$, respectively. It can be shown that the instantaneous frequency of the correlator output will be:

$$y_2(t) \approx \frac{\omega'}{2\pi} + A\omega_m(t_d + t') \cos\left(\omega_m t - \frac{\omega_m t_d - \omega_m t'}{2}\right). \tag{10}$$

Correlation of $y_2(t)$ with $\cos(\omega_m t)$ and $\sin(\omega_m t)$ over the observation duration T will give $t_{2,c}$ and $t_{2,s}$ as follows:

$$\begin{aligned} t_{2,c} &= \frac{2}{T}\int_0^T y_2(t)\ \cos(\omega_m t)\ dt \\ t_{2,s} &= \frac{2}{T}\int_0^T y_2(t)\ \sin(\omega_m t)\ dt \end{aligned} \tag{11}$$

and, as before, $t_{2,c}$ and $t_{2,s}$ will allow the calculation of $t_2 = t_d + t'$. Lastly, the true propagation delay can be computed as:

$$\hat{t}_d = \frac{t_1 + t_2}{2}. \tag{12}$$

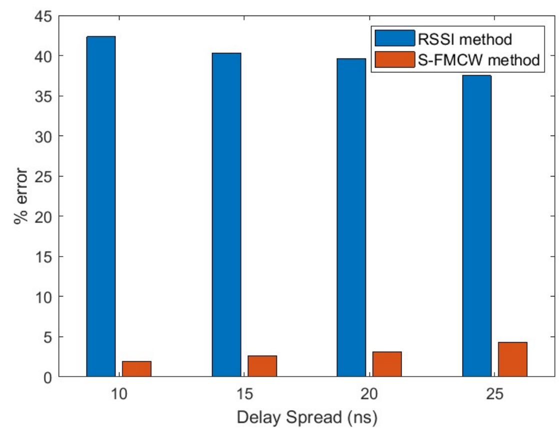

Figure 7: Comparison with RSSI based localization at the true distance of $d = 15$ meters.

4 Simulation Results

Simulations were performed to evaluate the ATS based S-FMCW technique For simulations, we use the exponential power delay profile channel model [5] that is widely used for indoor communications. According to this model, channel impulse response paths h_k are complex Gaussian random variables with $E[|h_k|^2 = ae^{-b(t_k - t_d)}$, where, b is a constant inversely related to the *Delay Spread* of the channel, a is an average power normalization constant and t_k is the delay of the specific path. The earliest path delay is considered to be the true range t_d of the devices. Transmit powers are set based on the assumption that the communication range of the devices is 30 meters. Signals are received in the presence of additive white Gaussian noise with a constant power spectral density.

Note: Bluetooth devices have been known to successfully communicate over distances of 200 meters or even more. But these are in unobstructed environments with only free space propagation losses. In this paper, the range is set to 30 meters due to indoor multipath fading effects being considered.

Figure 5 shows the Signal to Noise Ratio (SNR) of the range estimate at different distances between the Initiator and Reflector. It can been seen that accurate ranging is possible nearly up to the communication range of the link. Bluetooth requires a minimum receiver input SNR of 12–14 dB for the GFSK demodulator to provide acceptable performance [1]. This is sufficient for accurate localization as well. Another notable feature of FBR is that ranging accuracy does not drop rapidly with distance. Between 5 m and 20 m (a factor of 4), input SNR drops by at least 11 dB with just free-space losses alone (more when path loss exponents > 2 are considered). But, the SNR of the range estimate drops by only 6 dB. This can be attributed to the proportional relationship between the beat frequency and delay as can be seen from (5). Longer distances experience higher attenuation from propagation but also benefit from higher demodulation gains. This feature places FBR favorably when compared with either RSSI or PBR methods.

In Figure 6, the time needed to obtain 10 cm accuracy in ranging is shown for different distances between the Initiator and Reflector. Ten centimeter accuracy was chosen as it is considered necessary for many real world applications today [3]. With FBR, accuracy improves linearly with ATS duration due to (9) and (11). It can be seen that about 4 ms is sufficient to obtain this accuracy at a distance of 20m even with fading channels. The required observation time will be much smaller if there is a line-of-sight between the Reflector and Initiator. In Bluetooth, each time slot is 0.3125 ms long and packets may be 1, 3 or 5 slots long in asynchronous connection links. Figure 6 shows that longer packets may be needed to range targets further away. Alternatively, ranging accuracy may be relaxed with shorter packet durations.

In Figure 7, the ranging performance of FBR is compared with that of the RSSI method. Both FBR and RSSI methods can be applied to communication on a single Bluetooth 2 MHz channel. As expected, RSSI method is unreliable on fading channels without the use of finger-printing or other methods (not considered in this simulation). Performance of FBR gets slightly worse with increasing delay spread. The distance between the Initiator and Reflector was set to 15 m for this simulation. This is equivalent to $t_d = d/c \approx 48$ ns. The first path was set to this delay and other paths were randomly generated based on the exponential power-delay profile channel model. Performance of RSSI based ranging does not change much with the delay spread but gets worse as the actual distance between Initiator and Reflector is increased.

Comparison with PBR. PBR and FBR have qualitative similarities and differences. Both are based on converting time differences into angle differences— phase in PBR and frequency in FBR. Both are relatively immune to flat-fading multipath since the delays are derived from the phase or frequency rather than amplitude of the signal. However, a key difference is that frequency in FBR increases monotonically with delay whereas phase in PBR wraps around every 2π radians. To resolve the phase ambiguity, PBR measurements are required from multiple channels. PBR therefore requires a much wider bandwidth than FBR. Time needed for ranging is also higher with PBR since devices have to wait for the inter frame space time T_{IFS} to hop between Bluetooth channels [25]. Lastly, measurements over multiple channels increases the possibility of oscillator drift impacting PBR measurements. On the other hand, FBR is immune to oscillator mismatches due to the averaging in (9) and (11).

5 Conclusions

In this paper, a new method to accurately estimate the distance of a Bluetooth Reflector from an Initiator is described. In some ways, FBR is better than PBR in that transmissions on a single 2 MHz Bluetooth channel are sufficient to unambiguously determine the range. Furthermore, FBR signal can be generated in a way that is compliant with Bluetooth packet structure and modulation. The receiver does not need to be modified either. Verification of S-FMCW with Bluetooth hardware is an avenue for future work. Detailed quantitative comparison between FBR and PBR in different environments is another interesting aspect for further evaluation.

The main limitation of FBR is that significantly higher received power levels are needed when compared with PBR. This is due to the very low deviation ratio [9], particularly at shorter distances, of the FM signal in (4). This higher SNR requirement can be met by (i) increasing transmitter power, (ii) increasing A (and consequently the bandwidth) or (iii) increasing observation duration.

References

[1] Sergio Benedetto and Ezio Biglieri. 1999. *Principles of Digital transmission : with wireless applications.* Kluwer Academic/Plenum Publishers, New York.

[2] Pepijn Boer, Jac Romme, Jochem Govers, and Guido Dolmans. 2020. Performance of High-Accuracy Phase-Based Ranging in Multipath Environments. In *2020 IEEE 91st Vehicular Technology Conference (VTC2020-Spring).* 1–5.

[3] Kapseok Chang, Keunyoung Kim, Sanguk Lee, Jae-Hean Kim, Junghoon Lee, Hyungju Kim, Young-Jo Ko, Ilgyu Kim, and Seung Chan Bang. 2023. Technical challenges and solutions for 10 cm-level positioning accuracy towards 6G. *ICT Express* 9, 3 (2023), 492–506.

[4] B. S. Goh, A. K. Mahamad, S. Saon, K. Isa, H. A. Ameen, M. A. Ahmadon, and S. Yamaguchi. 2020. IoT Based Indoor Locating System (ILS) using Bluetooth Low Energy (BLE). In *2020 IEEE International Conference on Consumer Electronics (ICCE).* 1–4.

[5] J. Hansen. 2003. An analytical calculation of power delay profile and delay spread with experimental verification. *IEEE Communications Letters* 7, 6 (2003), 257–259.

[6] Zhanjun Hao, Yifei Gao, Yangyang Tang, Yue Wang, Kai Fan, and Fenfang Li. 2025. FMCW-based contactless heart rate monitoring. *Scientific Reports* 15, 1 (2025).

[7] Charu Jain, Gundepudi V Surya Sashank, Venkateswaran. N, and S. Markkandan. 2021. Low-cost BLE based Indoor Localization using RSSI Fingerprinting and Machine Learning. In *2021 Sixth International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET).* 363–367.

[8] Konstantinos Kotrotsios and Theofanis Orphanoudakis. 2021. Accurate Gridless Indoor Localization Based on Multiple Bluetooth Beacons and Machine Learning. In *2021 7th International Conference on Automation, Robotics and Applications (ICARA).* 190–194.

[9] B. P. Lathi and Z. Ding. 2018. *Modern Digital and Analog Communication* (fifth ed.). Oxford University Press.

[10] Noha Magdy, Mohannad Barakat, and Bassem Mokhtar. 2019. IoT Systems Internal Mapping using RTT with the integration of Blockchain technology. In *2019 Novel Intelligent and Leading Emerging Sciences Conference (NILES)*, Vol. 1. 154–157.

[11] Israel Martin-Escalona and Enrica Zola. 2020. Passive Round-Trip-Time Positioning in Dense IEEE 802.11 Networks. *Electronics* 9, 8 (2020). https://www.mdpi.com/2079-9292/9/8/1193

[12] Camellia S. Mouhammad, Ahmed Allam, Mohamed Abdel-Raouf, Ehab Shenouda, and Maha Elsabrouty. 2019. BLE Indoor Localization based on Improved RSSI and Trilateration. In *2019 7th International Japan-Africa Conference on Electronics, Communications, and Computations, (JAC-ECC).* 17–21.

[13] Santosh Nagaraj. 2025. On Mitigating the Mismatch Between FSK Tags and GFSK Receivers in BLE Backscatter. *IEEE Journal of Radio Frequency Identification* 9 (2025), 1–5.

[14] S. Nagaraj and C. Prasad. 2022. Sine-Cosine Modulation for SSB backscatter applications. *IEEE J. Radio Frequency Identification* (March 2022), 197–201.

[15] S. Nagaraj and R. Yaqo. 2021. A Frequency Modulation technique for SNR improvement in backscatter radios. *IEEE Commun. Lett.* (Dec 2021), 3956–3959.

[16] J. Rosenthal and M. Reynolds. 2019. A 1.0-Mb/s 198-pJ/bit Bluetooth Low-Energy compatible single sideband backscatter uplink for the NeuroDisc brain computer interface. *IEEE Transactions on Microw. Theory and Tech.* (October 2019), 4015–4022.

[17] Pulakesh Roy, Santu Mondal, Arindam Biswas, and Rajib Banerjee. 2024. A Comprehensive Study on BLE Enabled IoT Based Tracking System (BITS) for Indoor Environment. In *2024 2nd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT).* 578–585.

[18] Marco Terán, Juan Aranda, Henry Carrillo, Diego Mendez, and Carlos Parra. 2017. IoT-based system for indoor location using bluetooth low energy. In *2017 IEEE Colombian Conference on Communications and Computing (COLCOM).* 1–6.

[19] Orlandino Testa, Renato Cicchetti, Stefano Pisa, Erika Pittella, and Emanuele Piuzzi. 2025. A Novel FMCW Radar Scheme with Millimeter Motion Detection Capabilities Suitable for Cardio-Respiratory Monitoring. *Sensors* 25, 9 (2025).

[20] Yixin Wang, Qiang Ye, Jie Cheng, and Lei Wang. 2015. RSSI-Based Bluetooth Indoor Localization. In *2015 11th International Conference on Mobile Ad-hoc and Sensor Networks (MSN).* 165–171.

[21] Chen Weidong, Xu Shanjia, Wang Dongjin, and Liu Falin. 2000. Range performance analysis in linear FMCW radar. In *ICMMT 2000. 2000 2nd International Conference on Microwave and Millimeter Wave Technology Proceedings (Cat. No.00EX364).* 654–657.

[22] Martin Woolley. 2021. *Bluetooth core specification version 5.0 feature enhancements*. Bluetooth.

[23] Martin Woolley. 2021. *Bluetooth Direction Finding: A Technical Overview*. Bluetooth.

[24] Rongpeng Xiong, Xinxin Feng, Haifeng Zheng, and Zhizhang David Chen. 2020. Linear FMCW Radar System for Accurate Indoor Localization and Trajectory Detection. In *2020 International Conference on Computing, Networking and Communications (ICNC)*. 741–745.

[25] Pouria Zand, Jac Romme, Jochem Govers, Frank Pasveer, and Guido Dolmans. 2019. A high-accuracy phase-based ranging solution with Bluetooth Low Energy (BLE). In *2019 IEEE Wireless Communications and Networking Conference (WCNC)*. 1–8.

[26] Leifu Zhou, Qihao Huang, Tingting Lang, and Yanqing Qiu. 2024. Nonlinear FMCW Laser Ranging System Based on Sinusoidal Frequency Modulation. In *2024 22nd International Conference on Optical Communications and Networks (ICOCN)*. 1–3.